# CRSP/TAQ merged database - v2021.12

This folder contains code to merge data obtained from TAQ with the CRSP database released in December 2021.

## Prerequisites

Satisfy the prerequisites in the folder `crsp/202112`. Generate the files `master.csv.gz`, `nbbo.csv.gz`, `taqms.csv.gz`, `taqs.csv.gz` in the folder `taq/202112`.

## Installation

Install the dependencies with renv:

```
renv::restore()
```

## Usage

Generate the following data files by running the corresponding R script.

### File `ctm_mth.csv.gz`

```
Rscript ctm_mth.R
```

Contains monthly CRSP/TAQ merged data. Bid-ask spread estimates are signed and they are the root mean square spread within the month. A value of 0.01 corresponds to a spread of 1%.

| Field | Type | Description | Data |
|---|---|---|---|
| KYPERMNO | INTEGER | Unique permanent security identification number assigned by CRSP to each security | CRSP |
| YYYYMM | INTEGER | Year-month in the format YYYYMM | CRSP |
| NUMTRD | INTEGER | Average number of trades during market hours | WRDS IID |
| HJ | NUMERIC | Effective spread pre-computed by WRDS as described in Holden and Jacobsen (2014) | WRDS IID |
| QM | NUMERIC | Effective spread computed using the quoted midpoint as in Holden and Jacobsen (2014). This is a replication of HJ. | Daily TAQ |
| QW | NUMERIC | Effective spread computed using the quoted weighted-midpoint as in Hagstromer (2021) | Daily TAQ |
| EDGEMS, EDGES | NUMERIC | Effective spread obtained with the efficient estimator described in this paper | Daily TAQ, Monthly TAQ |
| ARMS, ARS | NUMERIC | Effective spread obtained with the estimator described in Abdi & Ranaldo (2017) | Daily TAQ, Monthly TAQ |
| CSMS, CSS | NUMERIC | Effective spread obtained with the estimator described in Corwin & Shultz (2012) | Daily TAQ, Monthly TAQ |
| ROLLMS, ROLLS | NUMERIC | Effective spread obtained with the estimator described in Roll (1984) | Daily TAQ, Monthly TAQ |